

2016.0 RANGE ROVER (LG), 501-09

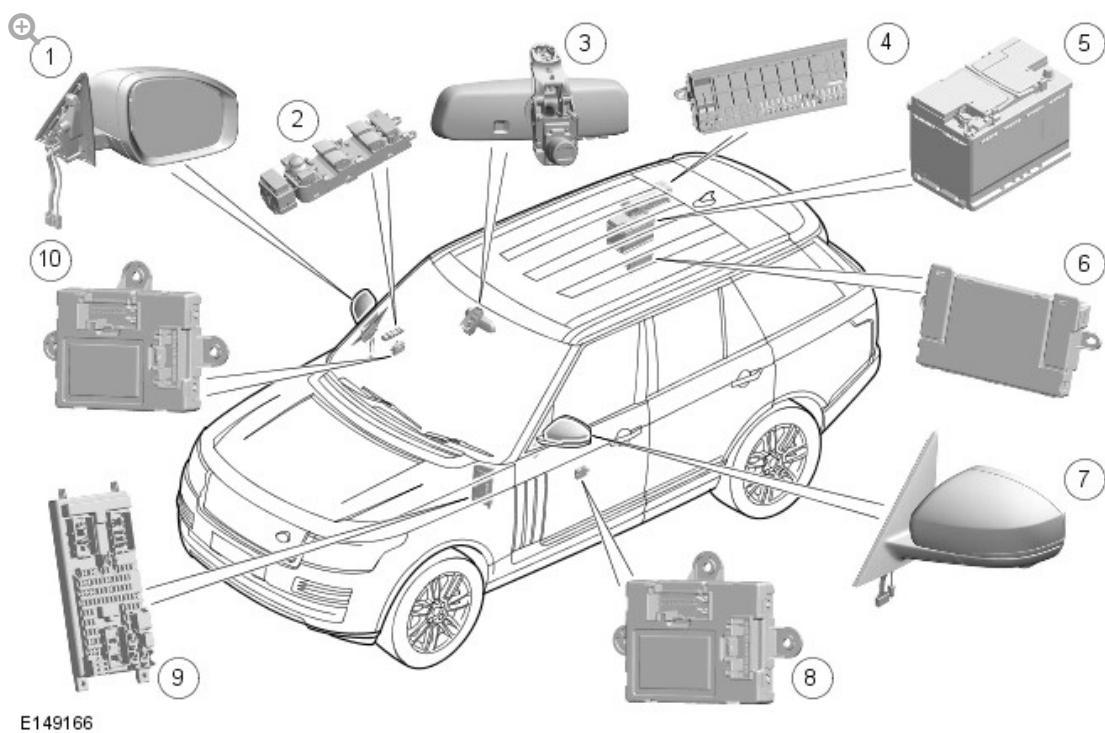
REAR VIEW MIRRORS

DESCRIPTION AND OPERATION

COMPONENT LOCATION

NOTE:

Right Hand Drive (RHD) shown, Left Hand Drive (LHD) similar



ITEM	DESCRIPTION
1	Right exterior mirror

2	Driver door switchpack
3	Interior rear view mirror
4	Rear Junction Box (RJB)
5	Battery
6	Battery Junction Box (BJB)
7	Left exterior mirror
8	Passenger Door Module (PDM)
9	Central Junction Box (CJB)
10	Driver Door Module (DDM)

OVERVIEW

Rear view mirrors consist of an interior mirror on the windshield and an exterior mirror on each front door cheater.

INTERIOR MIRROR

The interior rear view mirror is provided as an electrically operated automatic dimming type.

The automatic dimming mirror comprises an electro-chromatic glass housed within a surrounding case that is attached with a ball joint connector to the mirror stem.

Light sensors are mounted on the front and rear of the mirror surround case. The sensors control the automatic dimming feature to reduce glare from the headlights of following vehicles.

EXTERIOR MIRRORS

The exterior mirrors are bolted to the front door structure and incorporate the following features:

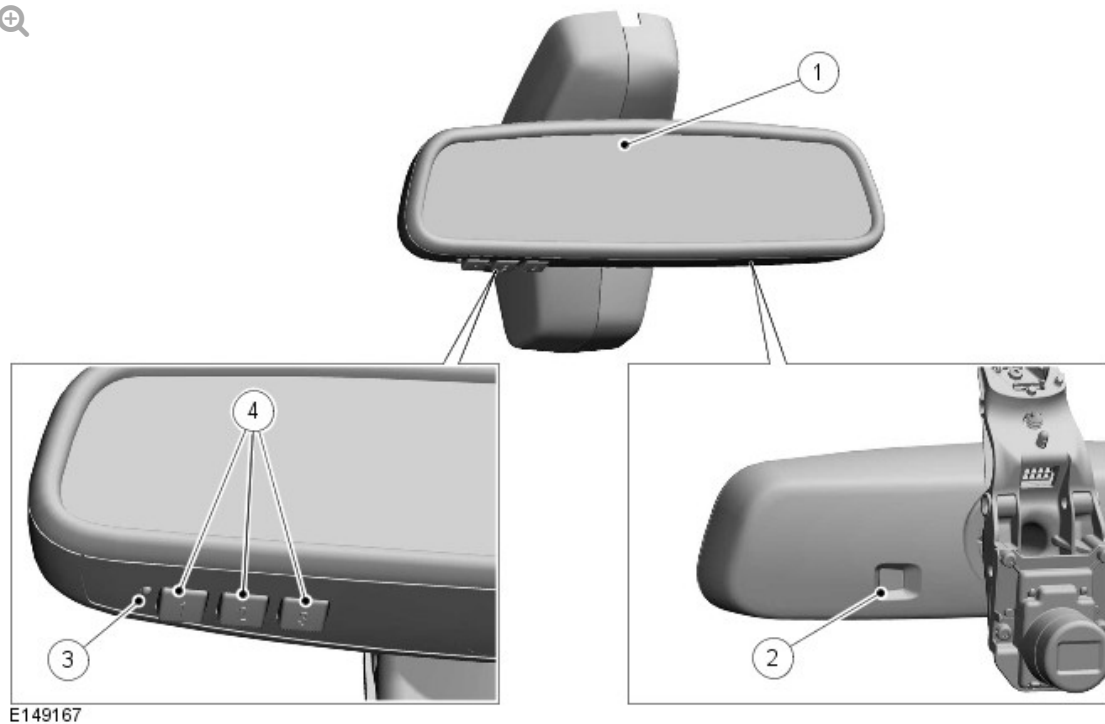
- Convex mirror glass.
- Electric adjustment.

- Mirror heating.
- Automatic dimming (model and market dependent).
- Memory recall and reverse dipping (model and market dependent).
- Mirror fold back function (model and market dependent).
- Approach lighting (model dependent).
For additional information, refer to: [Interior Lighting](#) (417-02 Interior Lighting, Description and Operation).
- Proximity cameras (model dependent).
- Ambient air temperature sensor.

Operation of the exterior mirrors is controlled by the 2 front door modules and the CJB. The door modules and the CJB communicate with each other. The door modules interpret the signals and control switch inputs into appropriate outputs for the exterior mirrors. For operation of the various exterior mirror functions the 2 door modules are powered by a permanent battery feed from the CJB.

DESCRIPTION

INTERIOR MIRROR



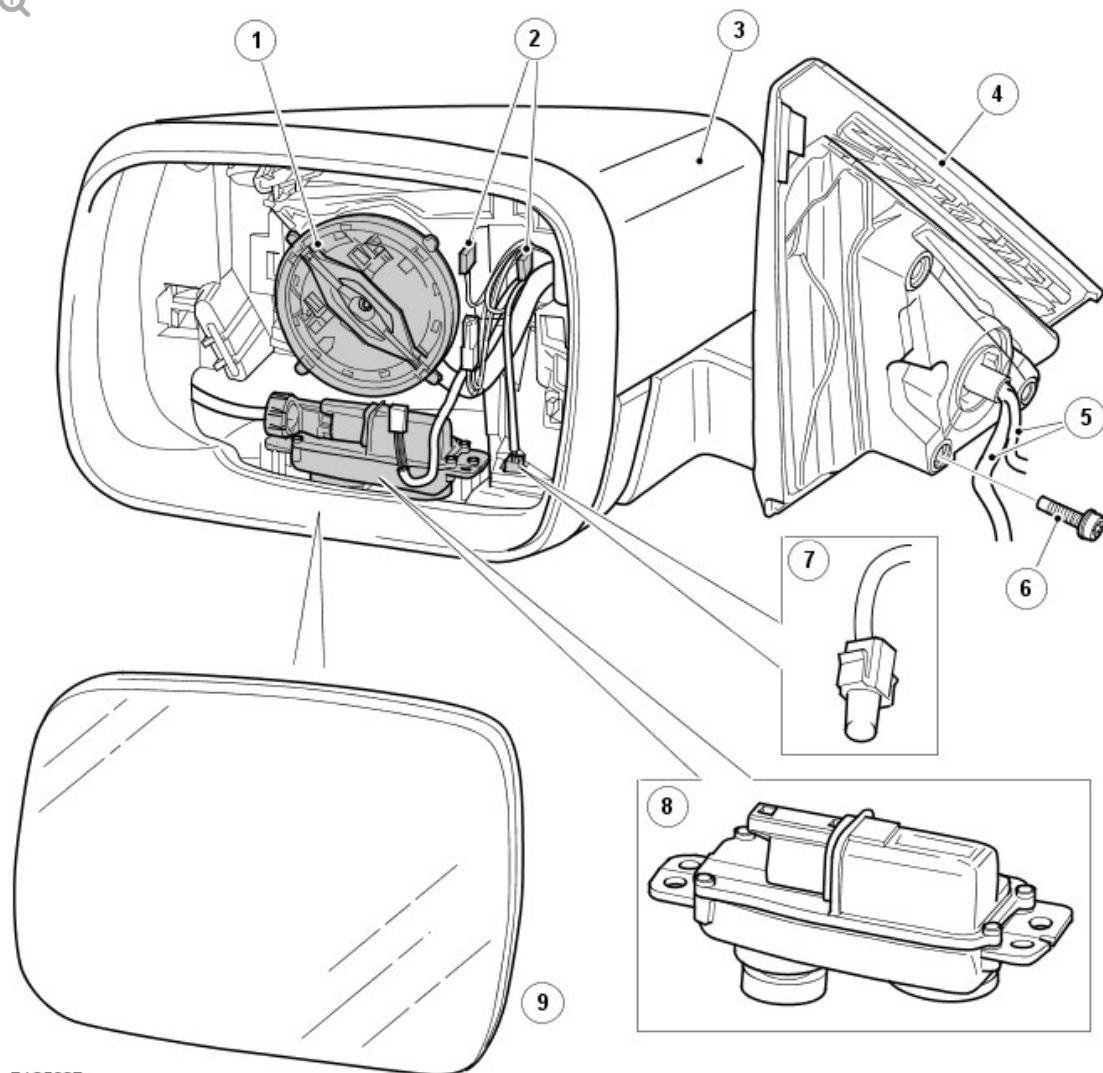
ITEM	DESCRIPTION
1	Rear light sensor
2	Front light sensor
3	Universal Transmitter Light Emitting Diode (LED) (market dependent
4	Universal Transmitter channel switch

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The automatic dimming mirror comprises an electro-chromatic glass housed within a surrounding case that is attached with a ball joint connector to the mirror stem.

Light sensors are mounted on the front and rear of the mirror surround case. The sensors control the automatic dimming feature to reduce glare from the headlights of following vehicles.

EXTERIOR MIRRORS



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ITEM	DESCRIPTION
1	Vertical and horizontal adjustment motor
2	Heating element connectors
3	Mirror body
4	Base frame
5	Fly leads (connected to door module)
6	Screw
7	Ambient Air Temperature (AAT) sensor
8	Proximity camera/Approach lamp
9	Mirror glass

HEATING

Heating of the exterior mirrors is controlled by the CJB and the respective door modules, and is active while the ignition is in power mode 6 or 7.

The CJB receives the ambient air temperature value from the ECM (engine control module) on the high speed CAN (controller area network) powertrain systems. The CJB converts the ambient air temperature value to an on-time percentage and transmits it on the medium speed CAN body systems to the two door modules, which then energize the exterior mirror heating elements accordingly. The on-time percentage is increased while the windshield wipers are on.

EXTERIOR MIRROR HEATING TIMES

Ambient Air Temperature in °C (°F)	< -10 (<14)	-10 to 0 (14 to 32)	0 to 15 (32 to 59)	15 to 25 (59 to 77)	25 to 35 (77 to 95)	> 35 (>95)
On-time Percentage	100%	75%	50%	25%	0%	0%
On-time Percentage With Wipers On	100%	100%	75%	50%	25%	0%

On vehicles with the parked heating function, exterior mirror heating may also operate when the parked heating function is active, depending on the ambient air temperature.

For additional information, refer to: [Auxiliary Climate Control](#) (412-02B Auxiliary Climate Control - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Description and Operation).

DIMMING

Exterior mirrors with automatic dimming are slaved to the interior mirror. The interior mirror determines the amount of dimming required and energizes the electrochromic elements in the exterior mirrors accordingly. Feed and ground wires from the interior mirror, for the electrochromic elements in the exterior mirrors, bypass the door modules and connect

directly to the exterior mirrors.

MIRROR MEMORY

The seat, exterior mirrors and steering column memory functions are integrated into the CJB.

The driver and front passenger door modules control the mirror memory while the driver seat module controls the seat and the CJB controls the steering column memory.

Located on the driver door trim panel, the memory switch and memory pre-set buttons can be used to store 3 different mirror positions. When a position is stored or recalled, the information is transmitted to the CJB via the medium speed CAN body systems. It is then relayed to the door modules. Each door module evaluates the recalling and storage commands transmitted via the medium speed CAN body systems for positions 1 to 3 and performs the necessary adjustments.

If a manual adjustment is selected while the mirror memory is operating, it will over-ride the memory recall function.

For mirror memory to operate, the mirror adjustment potentiometers must deliver a voltage value in the range of 80mV - 4.8V. Should a voltage applied be outside of this range the mirror will not operate when memory is selected.

REVERSE DIPPING

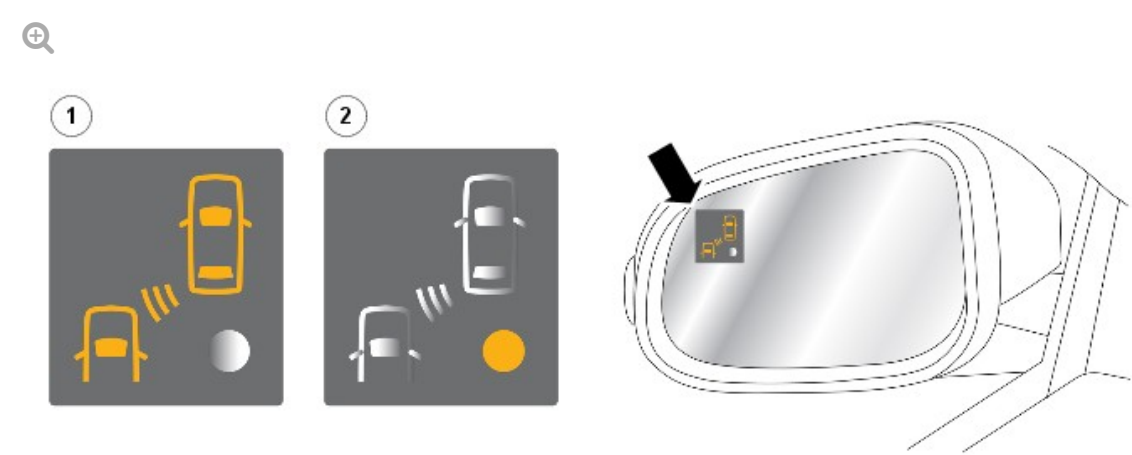
If the customer has activated reverse dipping when reverse gear is selected and the vehicle is in power mode 6 or 7, the driver and passenger exterior mirrors dip to provide a better view of the kerb. The mirrors return to their original position immediately upon reverse gear being disengaged.

The kerb view mirror position can be adjusted using the mirror adjustment multi-directional switch while reverse gear is selected and the vehicle in power mode 6 or 7.

BLIND SPOT MONITORING

NOTE:

Some variation in the illustrations may occur, but the essential information is always correct.



E97753

ITEM	DESCRIPTION
1	Warning alert icon
2	System status warning indicator

Blind spot monitoring function alerts the driver to a vehicle located in the vehicle blind spot. A warning indicator is located in each exterior mirror towards the outer edge.

For additional information, refer to: [Blindspot Monitoring System](#) (413-09A Warning Devices, Description and Operation).

ELECTROCHROMIC MIRRORS

The electrochromic mirrors automatically dim to reduce glare from the headlights of following vehicles in dark or low light conditions. In addition to dimming the interior mirror, the electrochromic mirror circuits also control the dimming of the two exterior mirrors, via power feed and ground

connections with the 2 exterior mirrors.

A light sensor on the front of the interior mirror monitors ambient light at the front of the vehicle and a light sensor in the interior mirror glass monitors the light coming from behind the vehicle. When the light from behind the vehicle exceeds the ambient light level, the electrochromic circuits simultaneously dim the interior and exterior mirrors. Dimming is inhibited when reverse gear is selected. The interior mirror is provided with a reverse gear signal by the Lighting Control Module (LCM).

OPERATION

AUTOMATIC DIMMING INTERIOR MIRROR

The automatic dimming function is permanently active when the ignition is in power mode 4 (Accessory) and power mode 6 (Ignition). The forward facing light sensor monitors the ambient light level at the front of the vehicle; the rearward facing light sensor monitors the light level coming from the rear of the vehicle. When light from the rear of the vehicle exceeds the ambient light level from the front of the vehicle, the automatic dimming circuit darkens the interior mirror surface.

Automatic dimming is inhibited when reverse gear is selected to provide the driver with maximum vision. The reverse gear signal is provided by the Transmission Control Module (TCM) via the high speed CAN powertrain systems to the CJB. The CJB then provides a power feed to the mirror.

Automatic dimming interior mirrors are connected to the vehicle wiring by an electrical connector in the mounting stem. The electrochromic mirror circuits and, where fitted, the universal transmitter, are powered by a feed from the CJB when the ignition is in power mode 6.

HOMELINK® UNIVERSAL TRANSMITTER

WARNING:

Before programming HomeLink® to a garage door opener or gate operator, make sure that people and objects are out of the way of the device to prevent potential harm or damage.

NOTES:

- Garage door openers manufactured after 1996 may be equipped with rolling code protection feature. If this is the case, you may need a stepladder or other sturdy, safe device to reach the 'Learn' or 'Smart' button on the garage door opener.
- When programming a garage door opener, it is advised to park outside of the garage.
- Additional HomeLink® information can be found at www.HomeLink.com. If additional programming support is needed, please call the toll-free HomeLink Hotline at 1-800-355-3515.
- Please note that the instructions below apply to the majority of HomeLink® use cases. However, there are some HomeLink® applications or HomeLink® compatible systems that require slightly different instructions. For more information and for custom training instructions specific to your device, visit <http://www.HomeLink.com>.
- HomeLink® is a registered trademark owned by GENTEX CORPORATION.

The HomeLink® wireless control system provides a convenient way to replace up to three hand-held Radio Frequency (RF) transmitters used to activate devices such as gate operators, garage door openers, entry door locks, security systems, even home lighting.

PROGRAMMING HOMELINK® UNIVERSAL TRANSMITTER

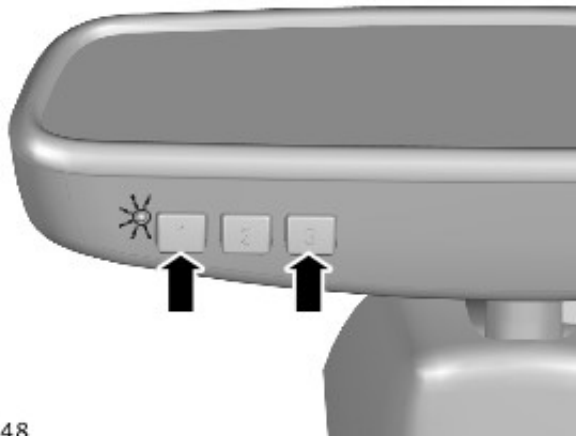
NOTE:

It is recommended that a new battery be placed in the hand-held transmitter of the device being programmed to HomeLink® for quicker training and accurate transmission of the Radio Frequency (RF) signal.

Step 1

Turn the ignition On.

Step 2



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Press and hold the outer two switches of the HomeLink® universal transmitter until the status indicator begins to flash.

Step 3

Release the switches.

This initializes the HomeLink® universal transmitter and erases previous settings from all three channels.

Step 4



E194850

Place the signal emitting end of the hand held transmitter 2 - 8 cm (1 - 3 inches) away against the underside of the rear view mirror. Keep the status indicator in view.

Simultaneously press and hold the activation switch on the hand held transmitter and the chosen switch of the HomeLink® universal transmitter. The status indicator begins to flash slowly.

NOTE:

Do not release the buttons until the following step has been completed.

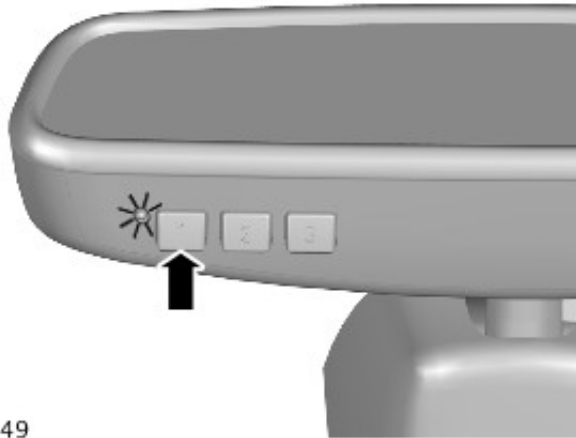
Step 5

The status indicator begins to flash rapidly. This can take up to 60 seconds.

Step 6

Release both switches.

Step 7



Press the programmed HomeLink® universal transmitter switch. The solid status indicator light indicates the successful programming.

To program another channel on the HomeLink® universal transmitter, repeat Steps 4 to 7.

Step 8

Turn the ignition Off.

To operate the previously programmed device, press and hold the chosen switch of the HomeLink® universal transmitter. Release the switch when the device begins to operate.

PROGRAMMING HOMELINK® UNIVERSAL TRANSMITTER - DEVICES WITH CODE PROTECTION FEATURE

The Radio Frequency (RF) signals used to operate some home/office systems incorporate a code protection feature. After a channel has been programmed from the hand held transmitter, these systems will need to be trained to accept the signal from the HomeLink® universal transmitter. To check if a system is code protected, operate the appropriate HomeLink® universal transmitter switch. If the status indicator flashes rapidly for 1 to 2

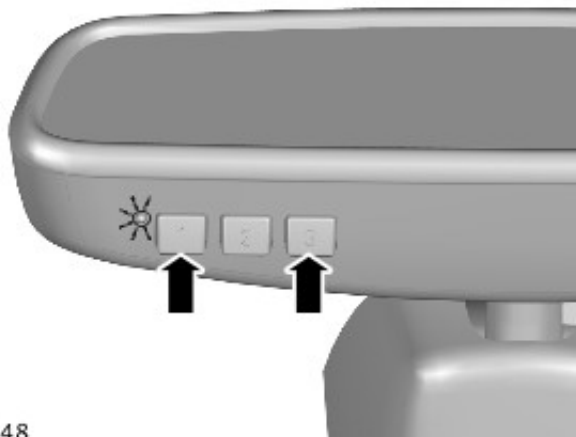
seconds before illuminating permanently, the system has a code protection feature.

Locate the 'Learn' or 'Smart' button at the garage door opener receiver (motor-head unit) in the garage before performing the following steps. This can usually be found directly on the motor-head unit (see the Garage Door Opener manual to identify the 'Learn Button').

Step 1

Turn the ignition On.

Step 2



E194848

Press and hold the outer two switches of the HomeLink® universal transmitter until the status indicator begins to flash.

Step 3

Release the switches.

This initializes the HomeLink® universal transmitter and erases previous settings from all three channels.

Step 4



E194850

Place the signal emitting end of the hand held transmitter 2 - 8 cm (1 - 3 inches) away against the underside of the rear view mirror. Keep the status indicator in view.

Simultaneously press and hold the activation switch on the hand held transmitter and the chosen switch of the HomeLink® universal transmitter. The status indicator begins to flash slowly.

NOTE:

DO NOT release the buttons until the following step has been completed.

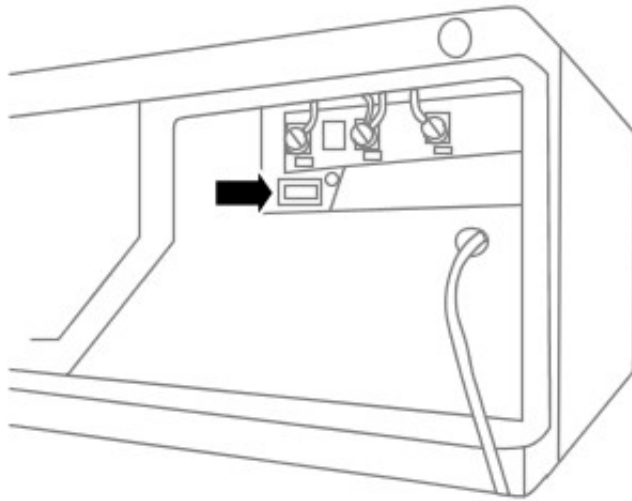
Step 5

The status indicator begins to flash rapidly. This can take up to 60 seconds.

Step 6

Release both switches.

Step 7



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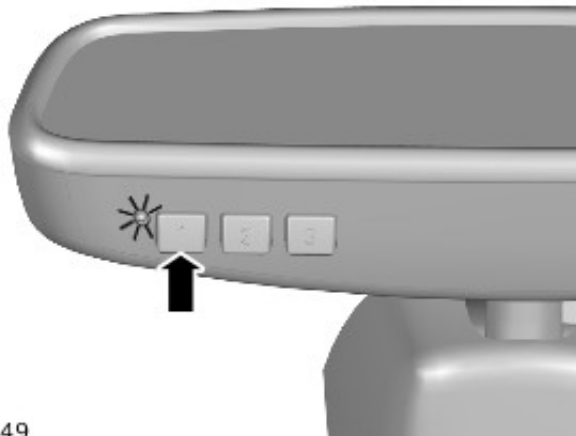
Press and release the 'Learn' or 'Smart' switch at the garage door opener receiver. The name and color of the switch may vary by manufacturer.

After the 'Learn' or 'Smart' switch has been released, a status indicator on the garage door opener receiver will continuously lit.

Step 8

NOTE:

Perform the following step within 30 seconds.



E194849

Return to the vehicle and press and hold the chosen switch of the universal transmitter for 3 seconds to activate the device. Repeat this step up to three times to complete the training.

If the device activates, programming is complete.

Step 9

Turn the ignition Off.

To operate the previously programmed device, press and hold the chosen switch of the universal transmitter. Release the switch when the device begins to operate.

AUTO HIGH BEAM

The auto high beam system is controlled by an auto high beam control module which is located in the interior rear view mirror body. The module and the CJB are connected via the medium speed CAN body systems.

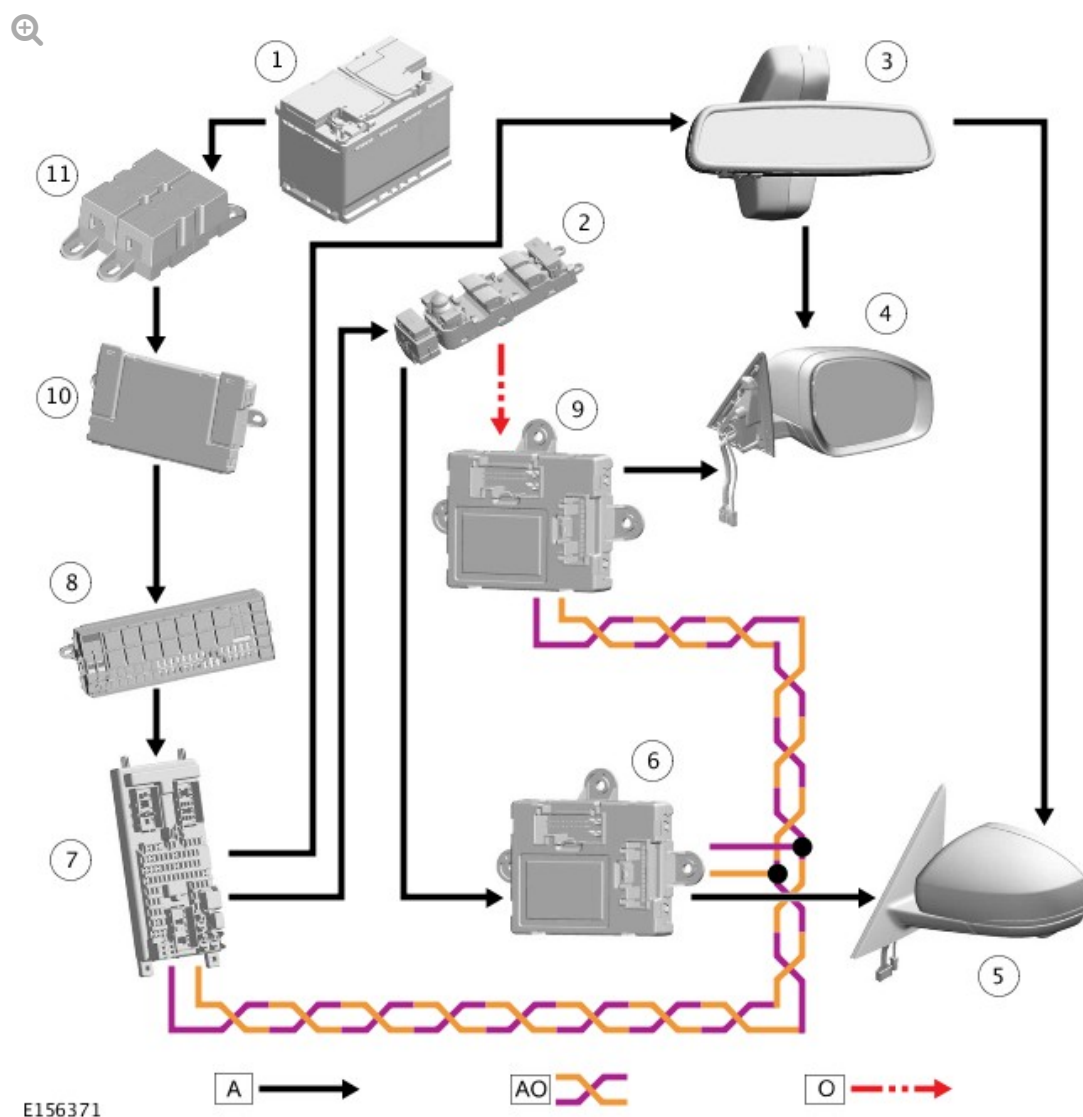
The auto high beam control module receives a power supply from the CJB when the ignition is in power mode 6 (ignition on). The rear view mirror also includes a low resolution camera (image) sensor which detects headlamps and tail lamps of preceding vehicles. The sensor is connected to the control module which evaluates the image data, checking for light intensity and

location.

If conditions are correct and auto high beam has been activated, the control module will activate the auto high beam by sending a high or low beam request message to the CJB via the medium speed CAN body systems. The CJB then controls the shutter in the Xenon projector module together with the high beam fill-in lamp.

For additional information, refer to: [Exterior Lighting](#) (417-01 Exterior Lighting, Description and Operation).

CONTROL DIAGRAM



A = HARDWIRED; AO = MEDIUM SPEED CAN BODY SYSTEMS; O = LIN (LOCAL INTERCONNECT NETWORK) BUS.

ITEM	DESCRIPTION
1	Battery
2	Driver door switchpack
3	Interior rear view mirror
4	Right exterior rear view mirror
5	Left exterior rear view mirror
6	Passenger Door Module (PDM)
7	Central Junction Box (CJB)
8	Rear Junction Box (RJB)
9	Driver Door Module (DDM)
10	Battery Junction Box (BJB)
11	Battery Junction Box (BJB) 2